

Pythagoras' theorem

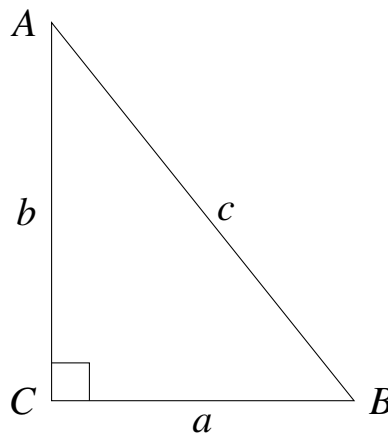
Covering

- Pythagoras' theorem

Pythagoras' theorem

The Pythagoras theorem is a fundamental principle in geometry that relates the sides of a right-angled triangle.

For a triangle ABC , with angle $C = 90^\circ$,



a and b are the legs, the shorter sides, and c is the hypotenuse, which is opposite to the right angle and is the longest side.

The basic integer solution for a , b and c is commonly $3^2 + 4^2 = 5^2$.

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Trigonometry

Covering

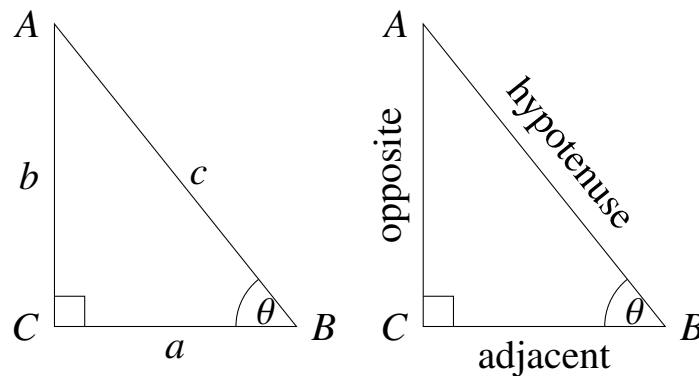
- Right-angled triangles
- Non-right-angled triangles
- Pythagoras' theorem and trigonometry in 3D

O levels and Extended

O levels and Extended

Basic Ratios

For a triangle ABC , with angle $C = 90^\circ$,



The three ratios are defined as the functions:

$$\begin{aligned} \sin \theta &= \frac{b}{c} = \frac{\text{opposite}}{\text{hypotenuse}} & \text{SOH} &= \text{Sin Opposite Hypotenuse} \\ \cos \theta &= \frac{a}{c} = \frac{\text{adjacent}}{\text{hypotenuse}} & \text{CAH} &= \text{Cosine Adjacent Hypotenuse} \\ \tan \theta &= \frac{a}{b} = \frac{\text{opposite}}{\text{adjacent}} & \text{TOA} &= \text{Tangent Opposite Adjacent} \end{aligned}$$

Obtuse Angles

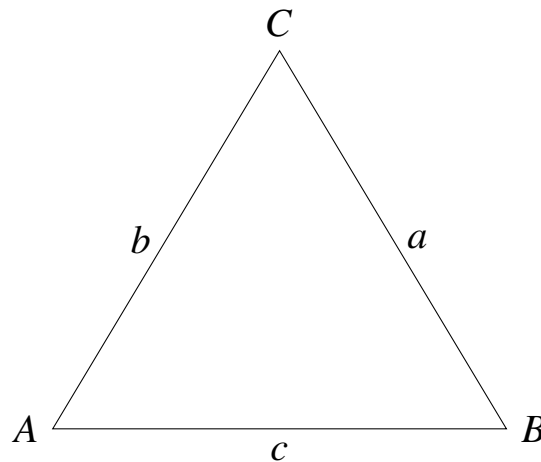
When the 3 trigonometric functions have a Obtuse angle rather than a Acute angle, we use:

$$\begin{aligned} \sin A &= \sin O \\ \cos A &= -\cos O \\ \tan A &= -\tan O \end{aligned}$$

Where A is acute angle and O is obtuse angle. $O = 180 - A \iff A = 180 - O$ can be used to find obtuse angles when needed.

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Sine Rule

Sine Rule is, ratios of side lengths to their opposite angles.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

This is applied when one pair, a side length and its opposite angle, is given.

Cosine Rule

Cosine rule is to find the length of a side using the 2 other sides and the angle they are making.

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = c^2 + a^2 - 2ca \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

For a as unknown side, b and c as other 2 sides. $\cos A$ is angle opposite to unknown side, or angle made by sides b and c .

We can use this to find the angle made by b and c instead if a is given as well.

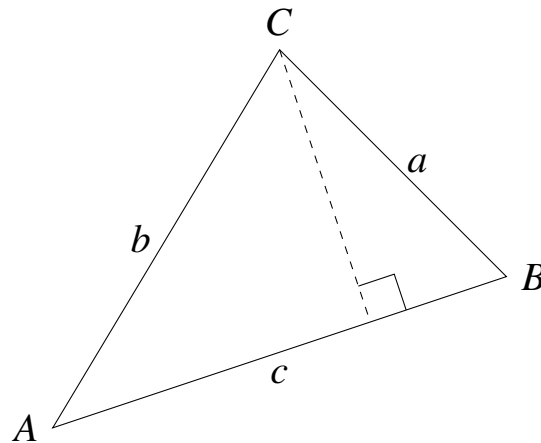
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc} \quad \cos B = \frac{c^2 + a^2 - b^2}{2ca} \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

All three formulas have a , b and c cycled, so same as the first, original, equation.

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Area of Triangle



To find the area of a triangle which doesn't have a right angle, we use:

$$A = \frac{1}{2}ab \sin C$$

For a , b as any 2 sides, and C as the angle they make.

This is to the fact if we make a perpendicular from the base AB to C , that is the minimum distance and hence the real height. The dashed line shows this.

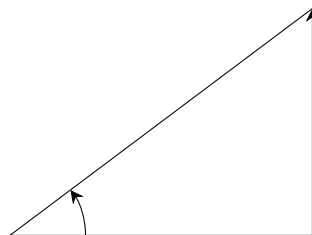
Angle of Elevation and Depression

O levels and Extended

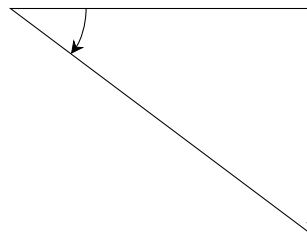
"Elevation" means height.

We measure this from a base line, e.g ground, and up to the wanted point.

Elevation



Depression



"Depression" means depth.

We measure this from a base line/high point and down to the wanted point.

Trigonometry in 3D

We need to be wary of how the diagram is, what the question demands and what/how a plane and line make a point/angle.

E.g finding the height of a pyramid, we measure from one vertex to the pyramid's apex, the angles measure from the base of the pyramid.

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Trig Values and Functions

Covering

- Exact trigonometric values
- Trigonometric functions

Extended

Extended

Trigonometric values

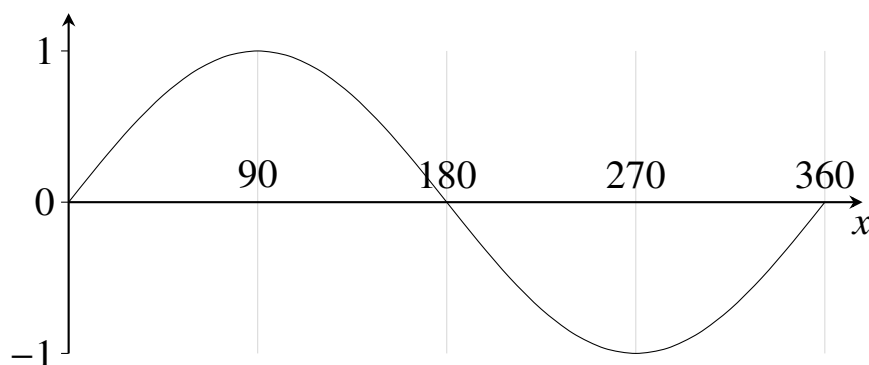
For this, a the table is good to see the values of $\sin x$, $\cos x$ and $\tan x$ for simple values of x .

x	0°	30°	45°	60°	90°
$\sin x$	$\frac{\sqrt{0}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{4}}{2}$
$\cos x$	$\frac{\sqrt{4}}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{1}}{2}$	$\frac{\sqrt{0}}{2}$
$\tan x$	$\frac{\sqrt{0}}{3}$	$\frac{\sqrt{3}}{3}$	$\frac{\sqrt{9}}{3}$	$\frac{\sqrt{27}}{3}$	undefined

x	0°	30°	45°	60°	90°
$\sin x$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos x$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan x$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Graphs and Solution of Trig Functions

The graph of a sine function is:

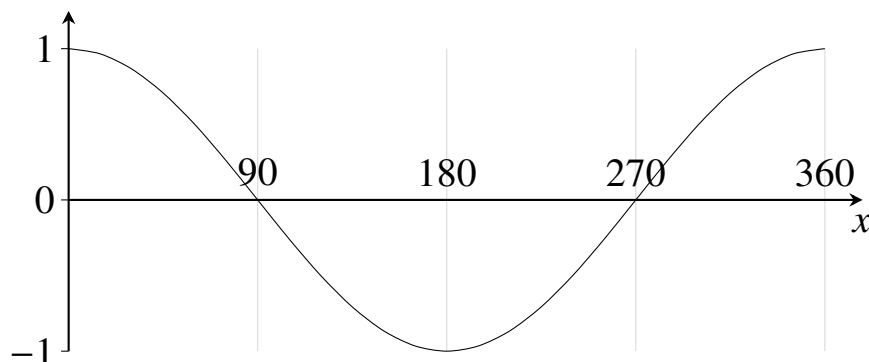


To solve $\sin x = c$, we firstly have $x = \sin^{-1} c$,
but there is a second solution $x = 180 - \sin^{-1} c$.

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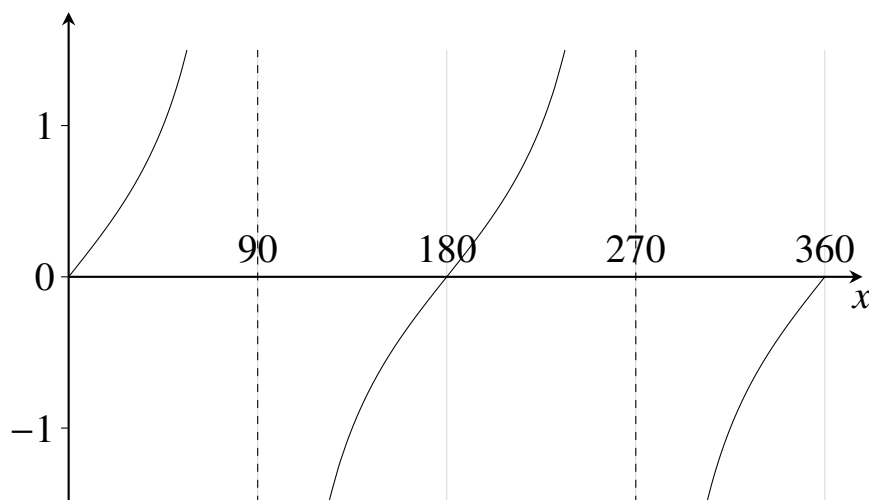


The graph of a cosine function is:



To solve $\cos x = c$, we firstly have $x = \cos^{-1} c$,
but there is a second solution $x = 360 - \cos^{-1} c$.

The graph of a tangent function is:



To solve $\tan x = c$, we firstly have $x = \tan^{-1} c$,
but there is a second solution $x = 180 + \tan^{-1} c$

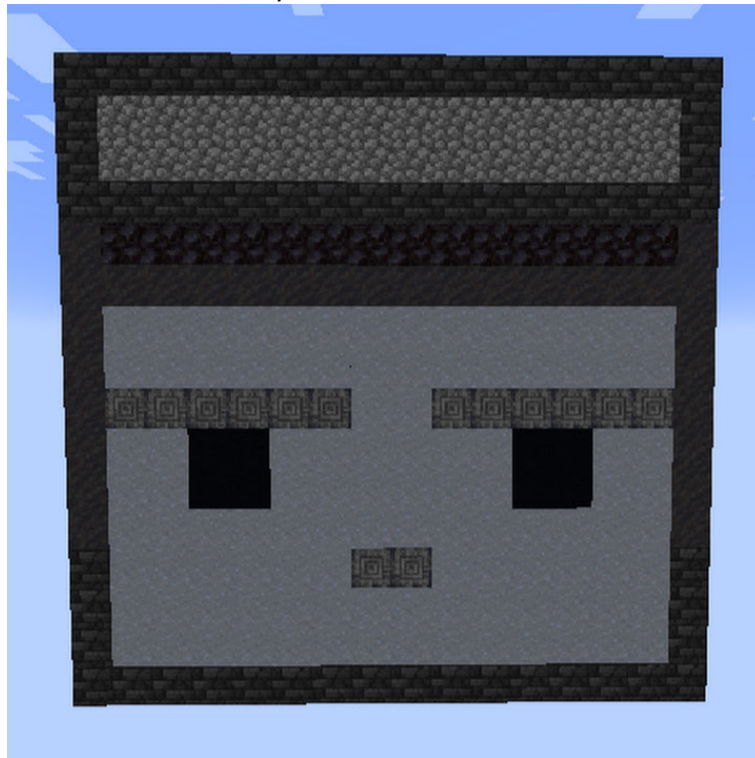


Mathematics is a beautiful subject—logical, creative, and deeply rewarding. It is truly worth investing your time in, not just for exams, but for the way it trains your mind to think clearly and solve problems with confidence. To excel in mathematics, you don't need a natural passion for the subject; what matters most is motivation and consistency. Progress comes from practice, patience, and the willingness to learn from mistakes. There will be challenges, but hard work, persistence, and never losing hope make all the difference. Keep trying, even when it feels difficult—every attempt strengthens your understanding. These notes are meant to reinforce that mindset and support your journey through **Mathematics O Levels and IGCSE**, reminding you that improvement is always possible with effort and determination.

If you don't know the method to study efficiently, try looking at the 67 Day Plan in the Study Material folder

[Here](#)

Checkout my channel @HarisRehmanGG



Where to find me:
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